

Requirements–Design Traceability Table

Point-of-Sale (POS) System

CSC–225 Software Engineering

1 Requirements–Design Traceability Table

This section presents the Requirements–Design Traceability Table for the Point-of-Sale (POS) System. The purpose of this table is to ensure that all functional requirements defined in the Software Requirements Specification (SRS) are fully addressed and mapped to the corresponding system design artifacts produced in Project Milestone 3.

Table 1: Requirements–Design Traceability Table

SRS Req. ID	Requirement Description	Use Case	DFD Process	Sequence Diagram	Activity Diagram
FR–01	User authentication with role–based access control	Login	User Authentication	Login Interaction	Login flow
FR–02	Order placement including table selection, menu items, discount, and complementary items	Place Order	Order Management	Order Placement Sequence	Order Placement
FR–03	Order processing, order ID generation, and status tracking	Modify / Cancel Order	Order Processing	Order Update Flow	Order Modification
FR–04	Kitchen operations including order acceptance and preparation	Prepare Order	Kitchen Processing	Kitchen Interaction	Order Preparation
FR–05	Order serving and order completion by waiter	Serve Order	Order Status Update	Serving Interaction	Order Completion
FR–06	Billing, payment processing, and receipt generation	Process Payment	Billing and Payment	Payment Sequence	Payment flow
FR–07	Customer feedback submission after payment	Provide Feedback	Feedback Processing	Feedback Submission	Feedback activity

SRS Req. ID	Requirement Description	Use Case	DFD Process	Sequence Diagram	Activi gram
FR-08	Administrative management of users, menu items, reports, and complementary items	Manage Sys-tem	Administration	Admin Opera-tions	Admin flow